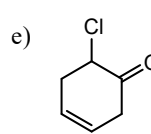
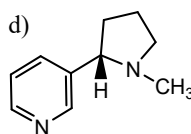
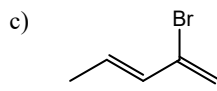
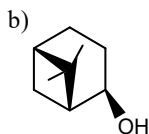
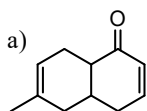
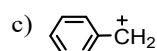
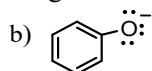
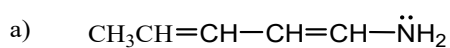


FOC work sheet -I

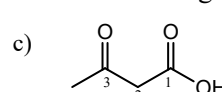
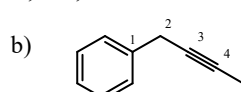
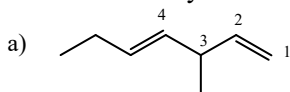
1 Find the molecular formula.



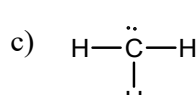
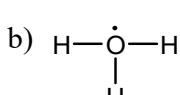
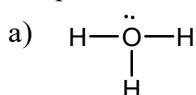
2 Write all resonance contributors for the following



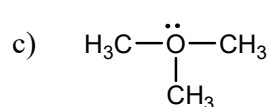
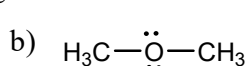
3 Determine the hybridization of C1, C2, C3 and C4 for each of the following compounds



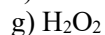
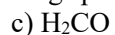
4 Determine the formal charge at all the atoms in each of the following species and the net charge on the species.



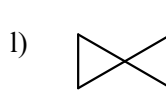
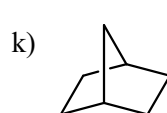
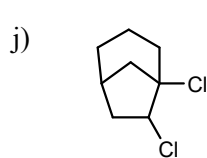
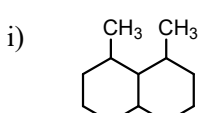
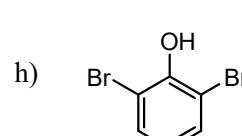
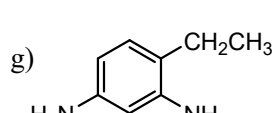
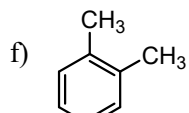
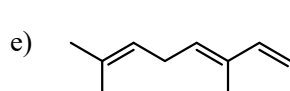
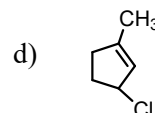
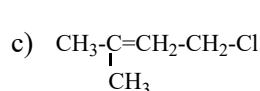
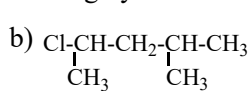
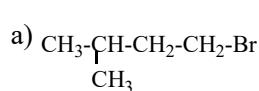
5 What is the formal charge of oxygen in each of the following Lewis structure?



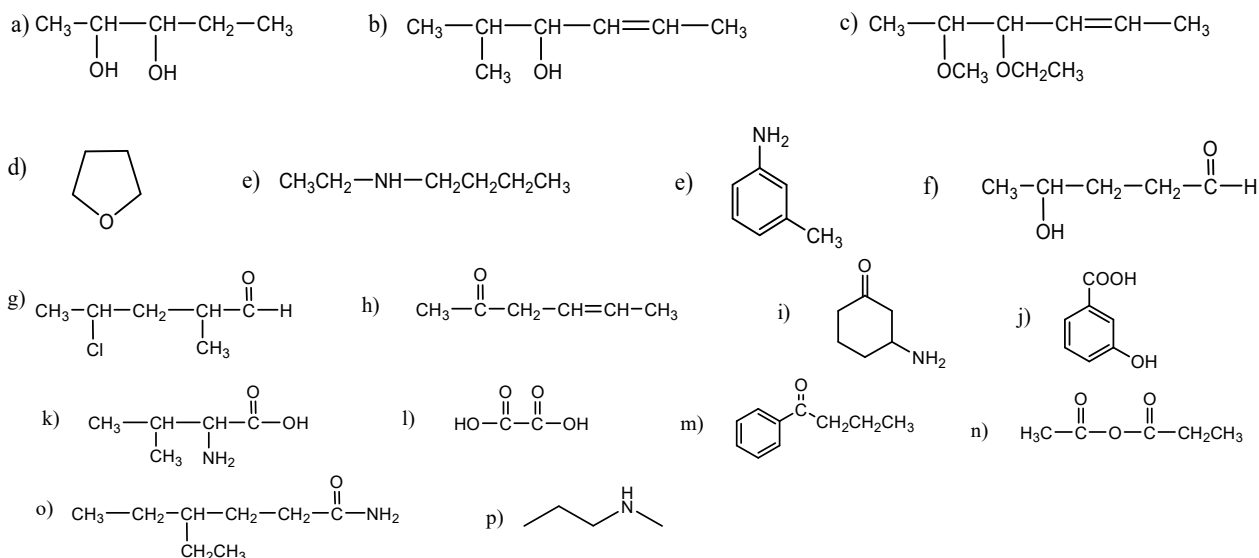
6. Draw the Lewis structure for the following species



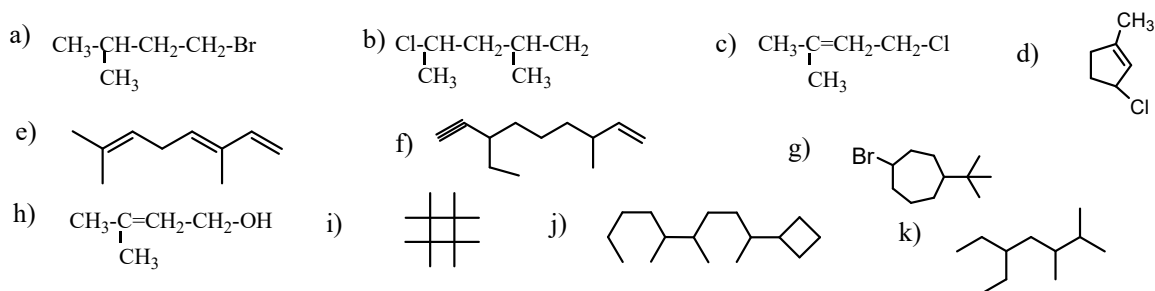
7. Write the **IUPAC** names of the following hydrocarbons



8. Write the **IUPAC** names of the following compounds



7. Name the following compounds according to IUPAC rule



8. Draw the structure of the following

- | | |
|--|-------------------------------------|
| a) 5-Chloro-4-iodo-6-methyl-1-heptyne | e) Diethyl ether |
| b) 6-tert-butyl-5-isopropyl-2,7-dimethylnonane | f) (Z)-3-Fluoro-2-methyl -3-hexene. |
| c) (E)-1-bromo-2-chloro-1-pentene | g) 1-ethoxy-1-methylcyclohexane |
| d) Bicyclo[2,2,2]octane | h) Bicyclo[5,2,0]nonane |

9. The name given below is **incorrect** IUPAC name for the hydrocarbon. Find the correct IUPAC name.

- | | |
|--|-----------------------------------|
| a) 1,2,4,4-tetramethylcyclohexane | d) 3-methylbicyclo[2,2,1]heptane |
| b) 3-sec-butylpentane | e) 2,2-diethylheptane |
| c) 1-cyclopropyl-3,3-dimethylcycloheptane. | f) 3,6-dipropylspiro[5,5]undecane |

10. Assign E or Z configuration to the following alkenes:

